

Table of Contents

Introduction.....	3
Methodology.....	3
Step 1.....	3
Step 2.....	5
Step 3.....	6
Step 4.....	7
Step 5.....	10
Step 6.....	12
References.....	15

Introduction

Portfolio optimization is part of investment management that enables investors to configure their portfolio's assets in a manner that maximizes long-term returns while considering the associated risks. The Kelly Criterion is one of the methods used in portfolio optimization that differs from traditional mean-variance approaches by maximizing the expected logarithmic growth rate of wealth, although it still relies on expected return and covariance estimates that are subject to estimation error. In the modern investment environment, machine learning techniques are increasingly used alongside the Kelly Criterion to improve the robustness of portfolio optimization by reducing estimation error in model inputs.

Methodology

A model will be built with Kelly Criterion. From the model, 3 Kelly strategies will be compared, namely Half-Kelly, Full-Kelly, and Double-Kelly. Then the model is backtesting in K-fold cross validation using Full-Kelly as benchmark.

Denoising, detoning, and clustering techniques to improve the model inputs are explored, implemented, and discussed in this project.

Step 1

Portfolio from group work project 1:

No.	Ticker	Company
1	TSLA	Tesla, Inc.
2	WMT	Walmart Inc.
3	BAC	Bank of America Corp
4	GS	The Goldman Sachs Group, Inc.
5	LLY	Eli Lilly and Company
6	MRK	Merck & Co., Inc.

7	GOOG	Alphabet Inc.
8	META	Meta Platforms, Inc.
9	AAPL	Apple Inc.
10	XOM	Exxon Mobil Corporation

Constrain:

- $\sum_i w_i = 1$
- $0 \leq w_i \leq 0.2$
- Data Period: January 1, 2024 - September 30, 2025

The optimal Kelly portfolio and constrained optimal Kelly portfolio after solving the Kelly objective are as below.

No.	Asset Ticker	Standard Asset Weight	Constrained Asset Weight
1	META	0.2235	0.2000
2	TSLA	0.2134	0.2000
3	GS	0.2064	0.2000
4	WMT	0.1738	0.1956
5	GOOG	0.1198	0.1340
6	BAC	0.0631	0.0704
7	LLY	0.0000	0.0000
8	MRK	0.0000	0.0000
9	AAPL	0.0000	0.0000
10	XOM	0.0000	0.0000

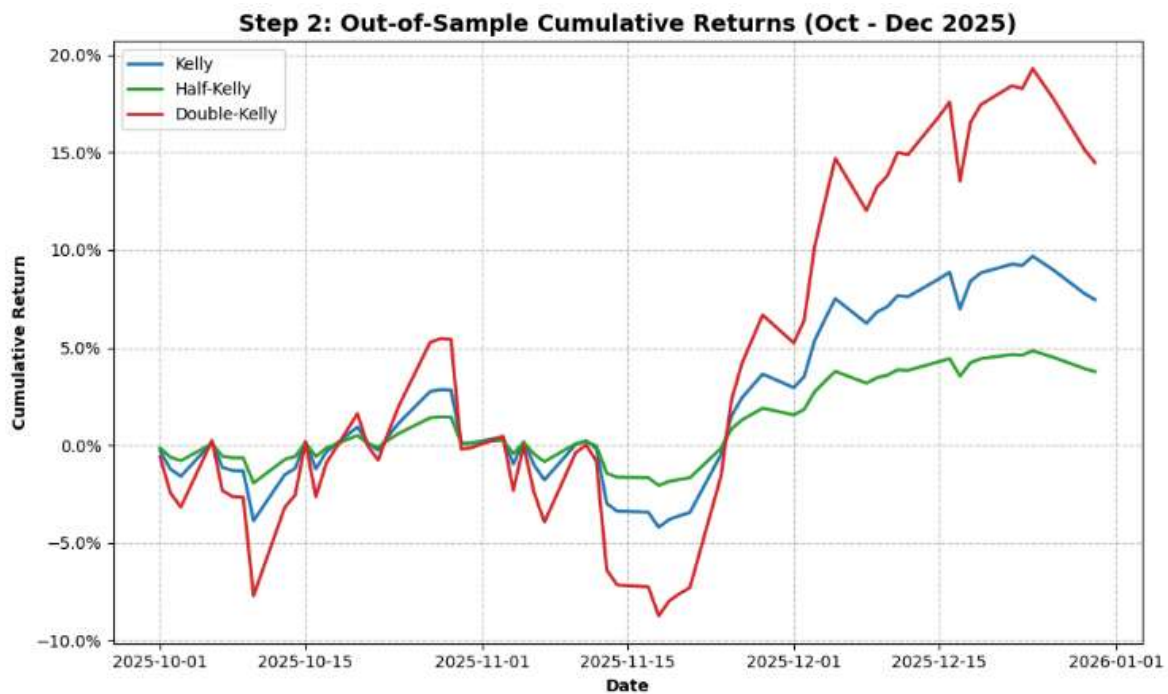
The maximization of the long-term return rate of the portfolio depends on both the expected returns and the covariance structure between assets. Therefore, the weight allocation tends to concentrate on assets

that provide the highest risk-adjusted returns potential rather than allocating some across all assets in the portfolio. For assets that are highly positively correlated, one of the assets will allocate less or sometimes zero allocation for risk reduction since log utility function penalise more for losses than gain, unless the returns gained could compensate for the risk.

Step 2

Comparison of full-Kelly, half-Kelly, and double-Kelly strategies on the portfolio in step 1 for data period from October 1, 2025 to December 31, 2025 are as below table, assuming zero risk-free rate.

Strategy	Cumulative Return	Sharpe Ratio	Maximum Drawdown	95%-cVaR
Half-Kelly	3.77%	1.65	-3.46%	-1.23%
Full-Kelly	7.46%	1.65	-6.86%	-2.45%
Double-Kelly	14.48%	1.65	-13.48%	-4.90%



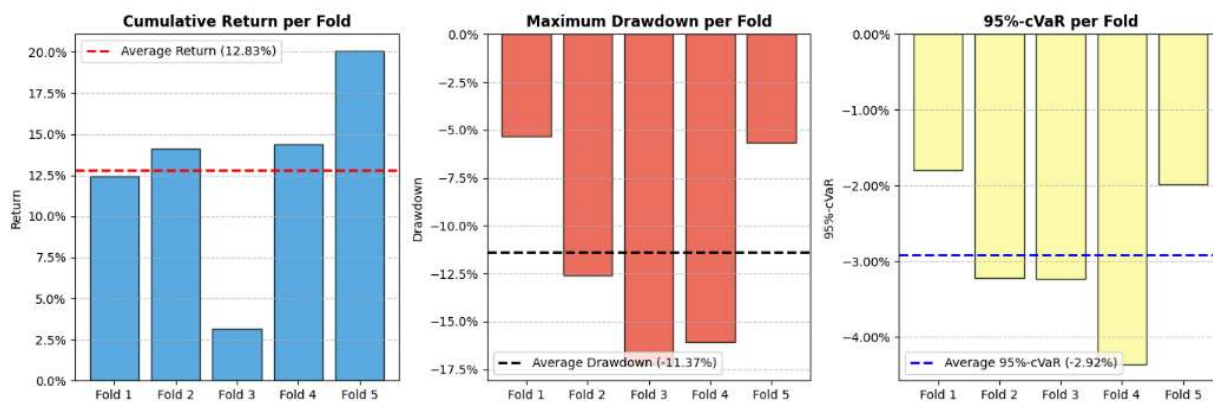
The results show that cumulative return, maximum drawdown, and 95% cVaR are about halved for

half-Kelly strategy and are approximately doubled for double-Kelly. It is approximate to a linear relationship to the scaling of full-Kelly. Note that the Sharpe ratio is not changed according to the scaling. The return per standard deviation is unchanged. This implies that there is a limitation to use Sharpe ratio alone to assess the leverage of Kelly portfolio as it does not capture the information/risk such as how the scaling/leverage affect the maximum drawdown and the 95% cVaR.

Step 3

To backtest the model using K-fold cross validation, the train-test data used are 2 years of data from January 1 2024 to December 31, 2025. The data will be split to 5 folds without shuffle to preserve the timeline order. Each fold will be used once as the test set and the rest will be the training set. Therefore, the train-test process will repeat 5 times. Finally, calculate the average across all folds to get the cross-validated expected performance.

Step 3: 5-Fold Cross Validation Results



No.	Asset Tickers	Full Kelly Weight					
		Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
1	GS	0.20	0.19	0.20	0.15	0.20	0.188
2	GOOG	0.16	0.20	0.20	0.19	0.10	0.170
3	WMT	0.15	0.13	0.15	0.15	0.20	0.156

No.	Asset Tickers	Full Kelly Weight					
		Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
4	TSLA	0.20	0.14	0.10	0.12	0.13	0.138
5	LLY	0.07	0.13	0.19	0.20	0.03	0.124
6	META	0.07	0.12	0.11	0.09	0.20	0.118
7	BAC	0.07	0.09	0.05	0.07	0.09	0.074
8	AAPL	0.07	0.00	0.00	0.04	0.04	0.030
9	XOM	0.00	0.00	0.00	0.00	0.01	0.002
10	MRK	0.00	0.00	0.00	0.00	0.00	0.000

The top 5 heavily weighted assets of the model are META, TSLA, GS, WMT, and GOOG, whereas the K-fold cross validation based on average weight are GS, GOOG, WMT, TSLA, and LLY. The assets chosen are mostly similar in both cases. The K-fold cross validation has an average return of 12.83%, a maximum drawdown of -11.37%, and 95%-cVaR of -2.92%, compared to the model with performance results of 7.46%, -6.86%, and -2.45% respectively.

Step 4

a) Improvements using denoising

The objective of denoising is to preserve informative signals by filtering out random noise in the correlation matrix. Noise often occurs due to small/limited sample size and random sampling fluctuations, which introduce estimation error (Timm 2021). Low noise level covariance matrix could improve the model robustness and thus, achieving more reliable results.

The eigenvalues of a purely random correlation matrix are expected to follow the Marcenko-Pastur distribution (MPD) (Pedersen 2021). Any eigenvalue that falls outside of the distribution is treated as a meaningful signal (especially those above upper bound) and retained during matrix reconstruction.

The denoising steps start from computing the correlation matrix. Then, Principal Component Analysis (PCA) is performed on the correlation matrix to obtain its eigenvalues and eigenvectors. These eigenvalues are then evaluated against MPD to filter out noise from the signal. Next, the noisy eigenvalues are averaged and replaced with the average value, while the signal eigenvalues are kept unchanged. After modifying the eigenvalues, the correlation matrix is reconstructed using the modified eigenvalues and the original eigenvectors to obtain the denoised correlation matrix. Finally, the denoised correlation matrix is converted to covariance matrix for portfolio optimization (Pedersen 2021).

Feature of denoising:

- I. Involve Principal Component Analysis
- II. Decompose the correlation matrix into eigenvalues and eigenvectors and reconstruct it afterwards
- III. Use of Marcenko-Pastur distribution to identify noise
- IV. Noisy eigenvalues are either replaced with their average value or shrunk

Benefit of denoising:

- I. Improved separation between informative factors and random fluctuations
- II. Reduce estimation error in covariance matrix
- III. More stable and robust portfolio allocation

b) Improvements using clustering (Hierarchical Risk Parity)

The Hierarchical Risk Parity (HRP) is a combination of hierarchical clustering algorithm and risk parity concept. Unlike denoising, HRP method avoids the unstable covariance matrix inversion that could be affected by estimation error caused by random fluctuations. The weight allocations are distributed to each cluster instead of directly distributed across all assets in the portfolio.

The correlation between assets is first calculated and converted into distances using the formula

$d_{ij} = \sqrt{\frac{1-\rho_{ij}}{2}}$. Next, hierarchical clustering is applied to segregate and group assets with similar behaviour into clusters. At this point, the hierarchy of groups is then reordered through Quasi-Diagonalization so that assets belonging to the same cluster appear next to each other and transform it to a usable matrix that reflects the hierarchy. Finally, the recursive bisection allocation method is used to split the portfolio into two clusters at a time (starting from the top of the dendrogram) and evaluate the variance of both clusters. According to the proportion of the variance contributed in each, higher variance cluster/asset will receive less weight allocated, while lower variance cluster/asset will have more weight allocated (Jain 2019).

Feature of HRP:

- I. Use of hierarchical clustering technique
- II. Introduced risk parity concept in determining weight allocation to assets
- III. Weight allocation to assets is not performed directly but starts from the top of the dendrogram

Benefit of HRP:

- I. Better interpretability with dendrogram
- II. More robust to estimation error because matrix inversion is not used
- III. Benefit of risk mitigation from diversification

c) Improvements using detoning

Detoning is actually an extended technique used after denoising to remove overall market factor from the correlation matrix. The overall market factor is common across all assets. Also, this factor risk is not diversifiable and could be dominant. As a result, other useful signals may be masked by the presence of the market factor (Timm 2021).

After obtaining the denoised correlation matrix, the market factor is extracted using PCA by decomposing the matrix into eigenvalues and eigenvectors. The largest eigenvalue and its corresponding eigenvector represent the dominant market factor. The market component is computed using the largest eigenvalue and its corresponding eigenvector and is then subtracted from the correlation matrix. Next, a scaling matrix is calculated using the diagonal value of the matrix for normalizing the correlation matrix after the market component removal. The scaling matrix normalizes the correlation matrix so that the diagonal value of the matrix becomes the value of 1. Finally, the normalized correlation matrix is converted to covariance matrix for portfolio optimization (Timm 2021).

Feature of detoning:

- I. Involve Principal Component Analysis
- II. Decompose the correlation matrix into eigenvalues and eigenvectors and reconstruct it for market component identification and removal
- III. Usually use after denoising the correlation matrix
- IV. Require scaling matrix to normalize the diagonal value of the correlation matrix back to value of 1 after component removal

Benefit of detoning:

- I. Allow optimization to focus on diversifiable components rather than the market risk
- II. Enhances useful signals and allows them to emerge more clearly
- III. Allow further refinement to correlation matrix after denoising

Step 5

To improve the results, we used the three ways of improving our portfolio which were stated in Step 4 on the 10-asset portfolio in Step 1: TSLA, WMT, BAC, GS, LLY, MRK, GOOG, META, AAPL, and XOM. As seen from the results of the original Step 1 portfolio, the use of Kelly optimization with constraints led

to a portfolio that heavily invested in META, TSLA, GS, WMT, GOOG, and BAC, whereas the remaining four assets had zero investment.

The main purpose of Step 5 was to test if our machine learning dependent improvement could make the portfolio more robust. The task measured three different approaches:

Portfolio Method	Description
Constrained Kelly	Original benchmark portfolio from Step 1
Standard HRP	Uses hierarchical clustering and risk parity without denoising/detoning
ML HRP	Uses denoising + detoning + HRP together

The resulting portfolio weights were:

Asset	Constrained Kelly	Standard HRP	ML HRP Denoised + Detoned
TSLA	0.2000	0.0223	0.0251
WMT	0.1956	0.2018	0.1530
BAC	0.0704	0.0655	0.1385
GS	0.2000	0.0499	0.1055
LLY	0.0000	0.0604	0.0548
MRK	0.0000	0.1243	0.1127
GOOG	0.1340	0.1069	0.0786
META	0.2000	0.0524	0.0757
AAPL	0.0000	0.1221	0.1126
XOM	0.0000	0.1944	0.1435

The important point is that the constrained Kelly strategy was very concentrated since four out of ten securities were assigned a zero weight. On the contrary, HRP and ML HRP generated a more diversified allocation. The reason is that the Kelly criterion allocates based on risk-adjusted growth in history, but at

the same time it is exposed to errors of estimations of both means and covariances. The HRP method overcomes this issue because it clusters portfolios, avoiding the direct matrix inversion.

In turn, the ML HRP allocated by first de-noising the correlation matrix, then detoning the market part and using hierarchical clustering. The theory-based process looked like:

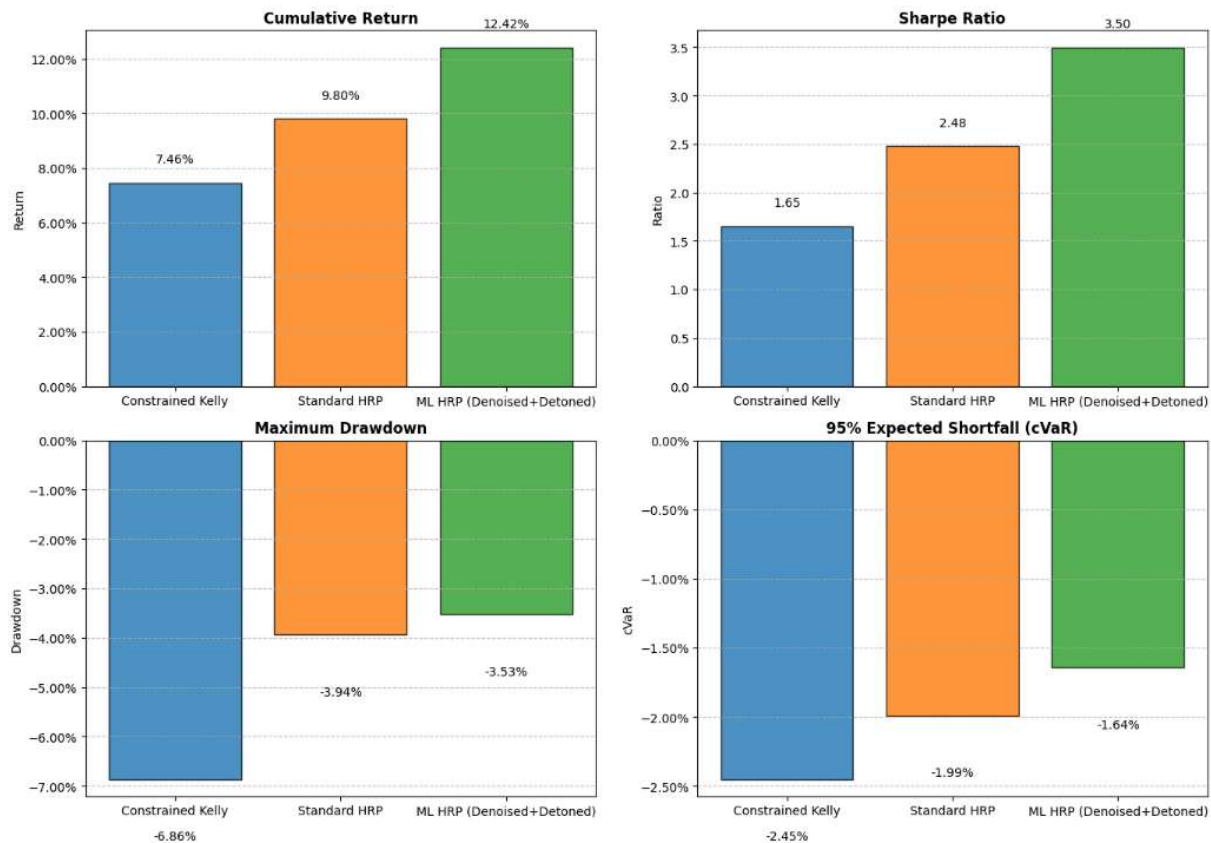
Raw returns → correlation matrix → denoising → detoning → HRP allocation

This process seems sensible since first the denoising process eliminates random correlation noise; secondly, detonation eliminates the broad market effect; thirdly, HRP utilizes the cleaned correlation pattern for capital allocation among clusters.

Step 6

The comparison of the three methods was carried out in the out-of-sample period from October 1, 2025, to December 31, 2025, based on the performance indicators: cumulative return, Sharpe ratio, maximum drawdown, 95% cVaR, and daily variance. This satisfies the project condition to compare whether multiple improvements are better than one improvement.

Strategy	Cumulative Return	Sharpe Ratio	Max Drawdown	95% cVaR	Daily Variance
Constrained Kelly	7.46%	1.65	-6.86%	-2.45%	0.000135
Standard HRP	9.80%	2.48	-3.94%	-1.99%	0.000096
ML HRP Denoised + Detoned	12.42%	3.50	-3.53%	-1.64%	0.000074



ML HRP provided the highest level of performance in terms of all five factors mentioned above – it gave the highest cumulative return, the highest Sharpe ratio, the lowest maximum drawdown, the least harmful 95% cVaR, and the lowest daily variance. That means that using several improvements was more efficient than a single improvement.

However, HRP performed better than the constrained Kelly model on its own. The cumulative return from HRP was 9.80% against 7.46% from constrained Kelly, while HRP had lower maximum drawdown at -3.94% against -6.86%. Hence, clustering alone was enough to reduce portfolio risk.

The most successful portfolio is the one where we combine denoising, detoning, and HRP. The cumulative return from ML HRP was 12.42%, which is better than 9.80% from HRP and 7.46% from the constrained Kelly model. In addition, it lowered daily variance from 0.000135 under Kelly to 0.000074.

It could be explained by the fact that the constrained Kelly approach included an investment in high growth and momentum stocks like TSLA, META, GS, and WMT. This strategy works well if these stocks have a positive trend, but its downside is the issue of concentration. The HRP method helped to address it because HRP spreads money among different stocks and groups. Furthermore, denoising and detoning helped HRP improve its correlation matrix so that it would not react sharply to random correlations.

An increase in complexity looks reasonable in this situation. Even though ML HRP demanded greater efforts from the point of view of its implementation, improvements were visible on all the selected metrics: higher returns, higher Sharpe ratio, lower drawdown, lower expected shortfall, and lower variance.

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